

Predicting pitcher ERAs in a given year using conventional stats

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Executive Summary

- **Purpose**

- This project is exploring which conventional pitching statistic can predict a pitcher's ERA in a given year and develops a statistical model predicting ERA, while discussing if using conventional stats is a good fit.

- **Data and Methods**

- The data set for this project was obtained from kaggle and has data from 2015-2024.
- Predictors examined: WHIP (Walks and Hits per Inning Pitched), HR.9 (Home Runs allowed per 9 Innings Pitched), K.9 (Strikeouts per 9 innings pitched), age, team, and season fixed effects.
- A fixed effects model was used to control for season-to-season variation.
- The model performance is evaluated by R-Squared, significance testing, and through comparison

- **Key Findings**

- WHIP is the strongest predictor of ERA, while HR.9, K.9, and team all have a statistically and practically significant impact on ERA as well.
- Age was statistically significant but had very small practical significance.
- Season fixed effects captured some unobserved variation.
- Team dichotomous variables were used for 4 statistically significant teams to account for more variation in ballpark and defense differences between teams.
- The final model explains a strong share of ERA variation, with an r-Squared score of .7888.

- **Implications**

- MLB teams can predict pitching performance reliably by using a small set of conventional metrics (WHIP, HR.9, K.9).
- The results can be compared with more advanced stats to see which is a better measure of predicting pitcher performance.

- **Conclusion**

- ERA can be predicted by using a small set of conventional statistics and fixed-effects modeling. The findings can provide guidance for player evaluation and resource allocation when signing players within professional baseball.

Introduction

Baseball is a very popular sport not only in America but in many Asian and Latin American countries as well. Its impact has spanned over 100 years and influences popular culture and linguistics. As America's past time, it is undoubtedly an important facet when looking at American culture throughout the years.

As the world has moved more digital and relied on computers more, all aspects of life have changed. Sports is no different in this regard. New types of stats and way of predicting performance have come along and have impacted sports front offices to re-evaluate how they evaluate players when signing and paying them. Of the 4 most popular sport leagues that exist in the US, Baseball is one of the pioneers in these analytics used by sports front offices.

Publications on analytics in baseball have existed since 1964 and sabermetrics in baseball have existed since the 1980s. These analyses and sabermetrics provide an empirical analysis into performance of each player and teams overall. Many of these saber metrics are not the typical statistics that people have historically used when looking at player performance like earned run average or batting average. The goal of these sabermetrics is to quantify more about the sport and provide more unbiased, mathematical ways to evaluate players. There are 4 main aspects that are looked at in sabermetrics, including hitting, pitching, base running, and fielding. However, despite existing for many years, there is still push back between if sabermetrics are a good tool to be using and if old school "eye tests" and traditional statistics are still sufficient in predicting player performance.

This project will focus on the metrics of pitching and attempt to explore the conventional statistics that can be used to predict earned run average (ERA), a main statistic for evaluating a pitchers performance, used by MLB front offices and other sports analysts. It will then discuss if using traditional statistics to predict pitchers performance is still a good tool, or if potentially using more advanced sabermetrics is better. We hypothesize that WHIP will have the most impact on ERA with age, team, strikeouts per 9 innings, and home runs per 9 innings all having a significant effect on ERA as well.

Literature Review

Since the idea of using regression and statistics has existed in baseball for many years, there are many other sources that have explored a similar topics. To narrow down the statistics to look at as the explanatory variables for ERA, the paper *Predicting Earned Run Average in Major League Baseball* by Paul C. Brendel was a valuable tool to look at. This paper goes into predicting ERA of pitchers using simple linear regression.

It looks at 4 different predictors of individual metrics for a pitcher, including strikeout to walk ratio, fastball velocity, first pitch strike percentage, and groundball to flyball percentage. These statistics are more about the individual pitcher's ability in a vacuum rather than their effect on opponent results. First pitch strike percentage is the pitcher's ability to throw a strike the first pitch of an at-bat and fastball velocity reflects how hard a given pitcher is able to throw. In this paper, it was discovered that of these 4 statistics, strikeout to walk ratio had the largest explanatory power of ERA with an R-Squared value of .32. The other three variables had very low R-Squared values, and even when added all together, the model's R-Squared value was only .36. This is an indication that model is not a very good fit for what it is attempting to do. Learning from this, instead of focusing on pitcher's individual abilities, the outcomes of opponents was noted as more impactful explanatory variables. Brendel explains that "K:BB may have been the best predictor since it was the only independent variable directly correlated with opponent on base results. Future studies may want to examine predictors that also have this quality, such as left on base percentage" (Brendel, n.d).

To address outcome based statistics, the next paper that was looked at was Beneventano, et al.'s paper *Predicting Run Production and Run Prevention in Baseball: The Impact of Sabermetrics*. This paper is attempting to compare whether sabermetrics are a good predictor of runs scored and run prevention in comparison to conventional statistics. The statistics for the explanatory variables were based more on opponents results and are as follows: Walks and Hits allowed per Inning Pitched (WHIP), Strikeouts per 9 innings, Home runs allowed per 9 innings, fielding percentage, and double plays turned. They also ran their model on sabermetric statistics, or statistics that are more advanced in calculation and may not make as much intuitive sense without the mathematical reasoning behind them. For their model, a stepwise-multiple-regression was used. The results of this model were promising, as their final model was able to get a very high R-Squared score of .988. They found that WHIP was the most important variable when predicting ERA, with the other variables still having significant impact. This paper was much more promising in its results as they had a very high R-Squared value. This led us to believe that opponent result statistics like WHIP and home runs per 9 innings were better predictors than individual pitching ability metrics. In this paper, they also examined the effect of defensive metrics like fielding percentage and double plays turned. These are stats that the pitcher is not directly responsible for, but rather the other 8 players on the field with them. This is an important thing to take note of and should not be ignored when making a model predicting ERA. One interesting note by the researchers is that none of the sabermetric statistics made it into their final model except WHIP, which was essentially just the combination of two conventional statistics, and is not as math based as some of the other sabermetric statistics. This meant that based on their findings that conventional statistics are still the best for determining pitcher performance.

The paper *Using the James-Stein Estimator to Predict Pitcher Performance* takes a very different approach to predicting ERA for a specific pitcher based on the previous year. The paper used a James-Stein estimator, which is a biased predictor that accounts for the pitcher's previous year performance and shrinks them back towards the group mean ERA. This brought the potential of biased results to the forefront and made us re-evaluate if this is something that needs to be accounted for in this project. While this is a valid concern, our project is aiming to predict a specific pitcher's ERA in a singular year and should not be affected by previous years in our calculation, so we did not adjust for it. However, the importance of normality testing was also discussed and made us make sure our errors were normally distributed.

Similar to how we focused on conventional statistics, *The Impact of Pitch Counts and Days of Rest on Performance Among Major-League Baseball Pitchers* by John C. Bradbury and Sean L. Forman examines the relationship between earned run average (ERA) and pitches per game, with predictions of performance based on one of the three scopes—the last game, the last five, or the last ten games pitched. In addition, this paper also looks at strikeout rate, home run rate, and walk rate separately against pitches per game, with ERA and home run rate having a positive relationship with pitches per game. However, ERA's relationship with other statistics provides a simple linear regression, and we sought to create a more complicated model that could incorporate multiple conventional statistics into our regression model.

On the contrary, *openWAR: An open source system for evaluating overall player performance in major league baseball* by Benjamin S. Baumer, Shane T. Jensen, and Gregory J. Matthews looks at more advanced statistics. Specifically, the literature looks at WAR (wins above replacement), a popular sabermetrics statistic that aggregates a baseball player's performance in pitching, fielding, hitting, and baserunning, and counters this statistic with openWAR, an alternative that builds off the aspects of WAR and consists of many different linear regressions that provide values for different aspects of a baseball player's performance. For pitching performance, a defensive run value, which is apportioned between pitchers and fielders, who are on the field to prevent the offense from getting runs. In cases where pitchers allow a batter to walk or make a home run, the entire defensive run value is assigned to the pitcher. Though the method for attributing and proportioning weight of responsibility to pitchers and fielders, it is much more difficult to measure, and requires many more steps to obtain a peek at the pitchers' performances.

Data and Identification Strategy

Data

The data for this project was obtained from a data set on kaggle. It has pitching data from 2015-2024 in both the hitting and pitching categories and got its data from publicly available sites like ESPN and fangraphs that have MLB data. Our variable of interest is earned run average and we are looking at the pitcher-year level.

The sample obtained for the project has 3141 observations. Each observation is a pitcher for one specific year that has at least 50 innings pitched. By limiting the data sets to pitchers with at least 50 innings pitched in a year, it allows for the pitchers who have very little innings and very outlier stats to be filtered out and not skew the results. In cases where a player plays for multiple teams in one season due to mid-season trades, the team is omitted for that player to avoid overestimating or underestimating the significance of a team.

For every observation at the pitcher-year level, we looked at WHIP (Walks and Hits per Inning Pitched), HR.9 (Home Runs allowed per 9 Innings Pitched), age of the player, K.9 (Strikeouts per 9 Innings Pitched), team the player played on, the season of the player's statistics, BB.9 (Walks per 9 Innings Pitched), batting average against a pitcher, shown as a rate of hits per at bat against the listed pitcher, and ERA (earned run average), which was our dependent variable.

Our initial theoretical model regresses ERA on WHIP. Our second model regresses ERA on WHIP, HR.9, K.9, and four teams that were used as dichotomous variables. The four teams included the Colorado Rockies, Chicago Cubs, Milwaukee Brewers, and Oakland Athletics, which were chosen by regressing ERA on WHIP, HR.9, K.9, and all 30 MLB teams, then picking out the four teams that displayed statistical significance. Our final model regresses ERA on WHIP, HR.9, K.9, the four selected teams from our second model, and includes an addition of fixed effects for season, indicated by year, to account for natural variation from year to year.

A regression was done on all the teams and the teams that were found to be statistically significant in explaining ERA were added to the mode. These were put in as dichotomous variables instead of a fixed effect on team because it was hypothesized that most teams would not have a significant effect on a pitcher's ERA, but some outlier teams would have some effect. This was also an attempt of capturing variability from each teams fielding ability and park factors, as these were not included in the data set. A better defense could lower a pitcher's ERA significantly and certain fields have different dimensions making it easier or harder to hit home runs. For example, Colorado's stadium is located in Denver, and with its elevation that

is much higher than every other ballpark in the league, it is easier to hit home runs there. This means a pitcher on Colorado, who will pitch a majority of their appearances at their home ballpark of Denver, will let up more home runs, on average, than their counterparts not on Colorado's team.

Results

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Table 1: Measuring the Relationship Between ERA and Conventional Pitching Statistics

	<i>Dependent variable:</i>	
	ERA	
	(1)	(2)
WHIP	4.713*** (0.063)	3.684*** (0.057)
HR.9		1.029*** (0.023)
K.9		-0.028*** (0.005)
CHC		0.196*** (0.057)
COL		-0.160*** (0.060)
MIL		-0.120** (0.056)
OAK		0.165*** (0.058)
Constant	-2.004*** (0.080)	-1.629*** (0.097)
Observations	3,141	3,141
R ²	0.644	0.788
Adjusted R ²	0.644	0.787
Residual Std. Error	0.707 (df = 3139)	0.546 (df = 3133)
F Statistic	5,671.500*** (df = 1; 3139)	1,662.744*** (df = 7; 3133)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 2: Measuring the Relationship Between ERA and Conventional Pitching Statistics with Season Fixed Effects

Dependent Variable:	ERA
Model:	Time FE Only (1)
<i>Variables</i>	
WHIP	3.691*** (0.0567)
HR.9	1.022*** (0.0234)
K.9	-0.0307*** (0.0054)
COL	0.1949*** (0.0569)
CHC	-0.1597*** (0.0598)
MIL	-0.1200** (0.0558)
OAK	0.1627*** (0.0583)
<i>Fixed-effects</i>	
Season	Yes
<i>Fit statistics</i>	
Observations	3,141
R ²	0.78941
Within R ²	0.78508
<i>IID standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

From the two tables seen, there are three different models. The first model in Table 1 is showing WHIP as a sole predictor of ERA. It's R-Squared value is .644, which is the highest of any individual statistic looked at. This one specific variable is already explaining 64.4% of the variability in ERA, which is a large amount.

The next model in Table 1 is when adding in the rest of the variables. As noted by the asterisks, each of these variables is statistically significant at the .05 level. Age was originally included with these other variables here, but was removed due to its small practical significance. As seen in Appendix A, the practical significance of age was very small. A one year change was only associated with a .2% of the mean change, thus this does not have much practical significance when trying to predict ERA. This model saw an increase in the R-Squared value, jumping to .788. This model's improved R-Squared value would make it a much better fit for predicting ERA than the first model, but there is still one model left to look at.

The model in Table 2 is the output from the model with the same parameters as the previous one, however it is controlling for fixed effects on season. It has an R-Squared value of .79, meaning 79% of the variation in ERA is explained by this model.

Our hypothesized equation looked like:

$$ERA_{it} = \beta_0 + \beta_1 WHIP_{it} + \beta_2 HR/9_{it} + \beta_3 K/9_{it} + \beta_4 Age_{it} + \beta_5 Team_i + \gamma_t + \varepsilon_{it}$$

Note that team is listed here, it will eventually turn into multiple dichotomous variables, but it was unknown at the beginning of the project which of the specific teams would have significance. So, team is a placeholder until that was discovered.

The final equation with the actual values for the coefficients ended up being:

$$\widehat{ERA}_{it} = 3.691 \cdot WHIP_{it} + 1.022 \cdot HR/9_{it} - .0307 \cdot K/9_{it} + .1949 \cdot COL_i - 0.1597 \cdot CHC_i + .1627 \cdot OAK_i - .12 \cdot MIL_i + \gamma_t + \varepsilon_{it}$$

Interpretation

Using the fixed-effects model seen in Table 2, the interpretation for each variable is as follows. For every additional walk or hit allowed in a inning pitched, it is associated with, on average, a 3.677 increase in ERA, controlling for all other variables and fixed effects on season. For each additional home run let up per 9 innings, it is associated with a 1.026 increase in ERA, on average, controlling for all other variables and fixed

effects on season. For each additional strikeout per 9 innings pitched, it is associated with a .0316 decrease in ERA, on average, controlling for all other variables and fixed effects on season. If a pitcher is on Colorado, it is associated with a .1905 ERA higher than a pitcher who is not on Colorado, on average, controlling for all other variables and fixed effects on season. If a pitcher is on the Chicago Cubs, it is associated with a .1508 ERA lower than someone who is not on the Chicago Cubs, on average, controlling for all other variables and fixed effects on season. If a pitcher is on Milwaukee, it is associated with a .1200 lower ERA, on average, than a pitcher who is not on Milwaukee, controlling for all other variables and fixed effects on season. If a pitcher is on Oakland, it is associated with a .1606 ERA increase than a pitcher not on Oakland, on average, controlling for all other variables and fixed effects on season. All of these variables are statistically significant at the .05 level.

Discussion Conclusions

Looking at each of the models, model 1 should not be looked at as a good fit model as it only takes into account one explanatory variable and has a lower R-Squared value than the other two. When choosing the best fit model from all that were run, model 3 seems to be the best fit. It has a high R-Squared score like model 2, however, it does take into account the variation from year to year that is seen in baseball and will do better at predicting a given pitcher's ERA in a given year. Year to year variation is an important factor to look at when examining any sport as rule changes, equipment changes, and culture changes all play in a role in how runs are scored. There is some speculation of the 2019 and 2017 seasons having "juiced baseballs", because the home run numbers from these year are significantly higher than any other year seen in the data. Years like this could skew simple multi-variate models, and so using fixed effects to capture this variation is useful.

One special thing to take note was that originally, walks per 9 innings and batting average against were also examined when doing this project. These were included in a model, however, they proved to show very high levels of multicollinearity with WHIP as shown in Appendix B. To address this, the model was redefined to only have WHIP. This one variable was chosen instead of the other two as it essentially captures the other two inside of it as they are both measuring the same thing and it would provide better, more interpretable results to someone looking at this model, as WHIP is a more commonly seen statistic than these other two rate based stats.

Another interesting thing when looking at this model is the fact that it meets the normality assumption. As seen in Appendix C, the histograms of the errors is fairly normal and when running the Anderson-Darling

test, it is also shown that the errors are normal. This was a surprising fact to us as baseball is a human-to-human interaction and categories are assigned to outcomes and it is not as simple as something like education level which has a set in stone definition. Score book keepers, players, and coaches all have a role in stats recorded in baseball and with this many people involved, some biases were expected to arise. But that was not the case as shown in Appendix C.

For future research, we could run a difference-in-difference design for the addition of the pitch timer in the MLB starting from the 2023 season. However, since our data set ran from 2015 to 2024, we did not have substantive data for a difference-in-difference model. In addition, whereas the 2023 season gave 20 seconds to pitchers between pitches when there were runners on bases, the MLB reduced the time limit to 18 seconds in the 2024 season. A difference-in-difference model would be more plausible three to five years in the future, when we have more seasons of data. Outside the MLB, which is what we focused on for this project, we could potentially apply the same methods to other professional baseball leagues, like Nippon Professional Baseball, which is based in Japan, and KBO, which is based in South Korea. At a smaller scale, we can also apply our regression model to college-level baseball, as leagues like the one in the United States under the National Collegiate Athletic Association (NCAA) and the one in Japan under the All Japan University Baseball Federation follow closely to their professional league counterparts in their respective countries.

With these results in mind, we advise MLB teams' general managers and coaches to focus on statistics like WHIP, home runs, and strikeouts when evaluating pitchers. This ranges from assessing a player's performance season to season and throughout the season to weighing the outcomes of roster changes, which could be a result of drafting new pitchers, trading for pitchers between and during seasons, or even signing free agent pitchers who are looking to join a team. These conventional statistics can also predict the future performance of a pitcher who is on the team.

References

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Appendix

A

Age

-0.009455953 [1] 3.981621 [1] 0.002373418 Age divided by the average value for ERA is .009/3.98 and gave us a value of .002 for the practical significance.

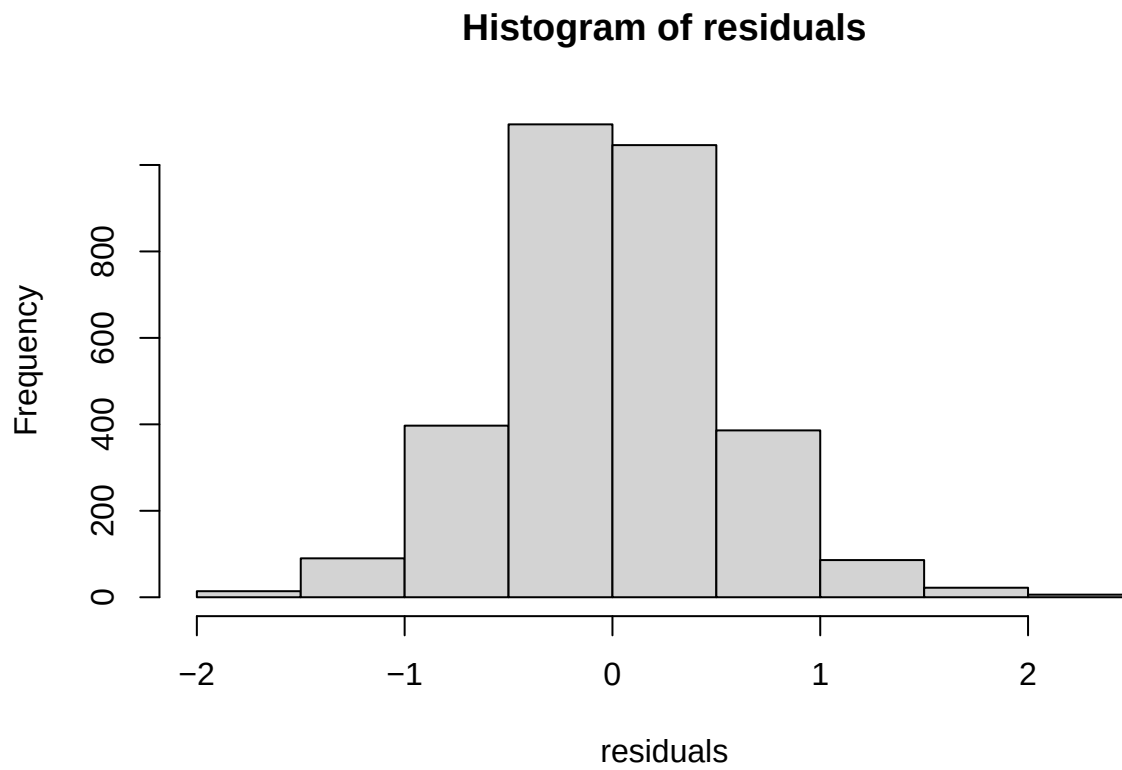
B

VIF scores for continuous variables with suspected multicollinearity. Walks per 9 and Batting Average Against were very correlated with WHIP.

##	WHIP	HR.9	K.9	Age	BB.9	AVG
##	144.342276	1.268922	1.770340	1.012805	43.881251	103.329146

C

Normality testing with Anderson-Darling test and histogram



```
##  
## Anderson-Darling normality test  
##  
## data: residuals  
## A = 3.7959, p-value = 1.839e-09
```